Single-Direction Refusal Across the Architectural Spectrum: Transformer, Hybrid, and Pure State-Space Models

Jordan F. McCann
Independent Researcher
jordanfmccann@gmail.com

May 10, 2026

Abstract

The single-direction refusal mechanism documented for transformer language models by Arditi et al. [1] raises a basic question: is the property attention-specific, or architecturally universal? We test it on three substrates — Llama-3.2-3B-Instruct (pure transformer, FP16), Zamba2-2.7B-Instruct (Mamba2 with $\sim 16\%$ attention hybrid, FP16), and Falcon-Mamba 7B Instruct (pure SSM, INT8) — with paired AdvBench/Alpaca prompts, mean-difference direction extraction, projection ablation, addition-induced refusal, and random-direction baselines. **Findings.** (1) On the **transformer baseline**, the canonical mechanism replicates: 5-seed mean ablation effectiveness 60.0% (95% CI $[51.4, 67.1]$); single-layer addition at $\alpha = \|r\|$ induces refusal in 78.6% of harmless prompts; capability preserved (MMLU 0.00 pp drop, HellaSwag 3.00 pp at the boundary). (2) On the **hybrid substrate**, the mechanism is the strongest in our panel: ablation refusal 0/22 (point effectiveness 100%; Wilson lower bound $\approx 64\%$); addition at $\alpha = 4\|r\|$ saturates at 24/24 (95% CI $[0.862, 1.000]$). (3) On the **pure-SSM substrate**, the representational signature is linearly perfect (Cohen's $d = 5.57$, ROC-AUC 1.000), but the INT8 substrate fails to follow instructions (baseline refusal 0/16); causal validation is inconclusive on this substrate. (4) The **architectural-spectrum hypothesis** (effectiveness $\propto$ attention fraction) is falsified in the *opposite* direction predicted on the two refusing substrates. (5) **Direction stability under instruction tuning** (Falcon-Mamba 7B base vs. Instruct cosine) is -0.006, with a chat-template-mismatch caveat. (6) Empirical comparison vs. a Sunku Mohan-style subspace-bottleneck yields a 14 pp point gap in our favor (bootstrap 95% CI $[0.0, 35.7]$ pp — suggestive but not significant at $N = 24$). We release bit-exact 9-extraction-point hooks for Mamba-1, Mamba-2 SSD, and Zamba2 hybrid architectures, and document a methodological-audit pattern that caught a Unicode-classifier bug inflating an earlier effectiveness number to a spurious 94.7%.

1 Introduction

The mechanistic interpretability of refusal in instruction-tuned language models is a load-bearing question for AI safety. Arditi et al. [1] showed that for transformer language models a *single direction* in residual-stream activation space mediates refusal: projection-based ablation of this direction reduces refusal rate on AdvBench-style harmful instructions from $> 80\%$ to $< 20\%$ with general capability preserved.

This raises a question the original paper did not address: **is the single-direction refusal mechanism architectural, or attention-specific?** State-space language models [3, 5] replace attention with a selective scan over hidden states. They share the same residual-stream substrate as transformers [4] but lack content-based attention as the primary mixing primitive. If the linear-representation hypothesis depends specifically on attention's content-based mixing, SSMs should not exhibit single-direction refusal; if it is about how instruction tuning concentrates safety signal in residual streams, SSMs should.

We pre-register four hypotheses with falsification thresholds (Section 2.4) and test them across a three-substrate panel: pure transformer (Llama-3.2-3B-Instruct), hybrid Mamba2 + attention (Zamba2-2.7B-Instruct), and pure SSM (Falcon-Mamba 7B Instruct, INT8). We report multi-seed effectiveness, capability preservation, cross-distribution generalization, multi-direction (PCA) analysis, instruction-tuning direction stability, and an empirical comparison against the methodologically adjacent Sunku Mohan et al. [11] subspace-bottleneck method.

1.1 Contributions

1. **Replication of the Arditi single-direction refusal mechanism on Llama-3.2-3B-Instruct** with full causal validation and 5-seed robustness CI: mean ablation effectiveness **60.0**% (95% CI $[51.4, 67.1]$); single-layer addition at $\alpha = \|r\|$ induces refusal in 78.6% of harmless prompts; random baseline 0%; Cohen's $d = 6.50$; ROC-AUC 1.000; capability preserved.

2. **First reported quantification of single-direction refusal on a hybrid SSM/attention model** (Zamba2-2.7B-Instruct, $\sim 16\%$ attention): ablation refusal rate 0/22 (point effectiveness 100%; Wilson lower bound $\approx 64\%$); addition at $\alpha = 4\|r\|$ at 24/24 (95% CI $[0.862, 1.000]$); Cohen's $d = 5.43$; ROC-AUC 1.000.

3. **First reported quantification of harmful-vs-harmless linear separability in a pure-SSM residual stream** (Falcon-Mamba 7B Instruct INT8 layer 32): Cohen's $d = 5.57$, ROC-AUC 1.000 — indistinguishable in linear-separability quality from transformer and hybrid baselines.

4. **H4 (effectiveness $\propto$ attention fraction) falsified in the opposite direction predicted** on the two refusing substrates: hybrid (16% attention) > pure transformer (100% attention). Caveat: 2-substrate spectrum, scale and instruction-tuning-procedure confounds.

5. **H2 (direction stability under instruction tuning) falsified on Falcon-Mamba 7B**: cosine $= -0.006$ between base and instruct directions at the same (layer, EP). Caveat: chat-template mismatch limits strength.

6. **Empirical comparison with Sunku Mohan et al. [11] subspace-bottleneck**: point gap of 14.3 pp in our method's favor on the same data and intervention type; 95% bootstrap CI on the gap is $[0.0, 35.7]$ pp — suggestive but not significant at $N = 24$.

7. **Methodological-audit pattern that caught a Unicode-classifier bug mid-execution** that inflated an earlier headline number to a false 94.7%; the corrected protocol yields the 60.0% multi-seed mean reported in (1). Released as a reusable verification-gate pattern.

8. **9-extraction-point hook infrastructure** for Mamba-1, Mamba-2 SSD, and Zamba2 hybrid layers — bit-exact verified (100/100 numerical-equivalence test on every substrate, max abs. diff. 0.0; spot-check $h_0 = \Delta_0 \cdot B_0 \cdot x_0$ bit-exact in fp32).

9. **Reproducibility envelope:** all experiments run on a single RTX 2060 12 GB. Full model SHAs (Section 2.1), pinned datasets (Section 2.7), seeds (Section 2.8), and one-command reproduction scripts.

1.2 What we do not claim

- **H1 confirmation or falsification on the pure-SSM side.** Falcon-Mamba 7B Instruct INT8 in our setup does not refuse harmful prompts at baseline (0/16), so the dependent variable is absent and causal-validation experiments cannot test H1 directly. The linear-separability finding is consistent with — but does not test — a single-direction mechanism in pure-SSM residual streams. The clean follow-up experiment is identified in Section 5.3.
- **The Phase 10 14 pp gap as statistically significant at $N = 24$.** Bootstrap 95% CI on the gap is $[0.0, 35.7]$ pp; a larger eval N is required to make the differentiation conclusive.
- **The H4 falsification as architecturally definitive.** Two refusing substrates is a small spectrum; scale (3B vs. 2.7B) and instruction-tuning-procedure differences are confounds.
- **The Phase 9 100% effectiveness as point-mass.** $N = 24$ makes the Wilson lower bound on effectiveness $\approx 64\%$; the result is "very strong" but not "exactly 100%."
- **Full-battery capability preservation.** Phase 5 used a 200-question subset (MMLU 100 + HellaSwag 100); subtle capability degradation in untested domains is not ruled out.

2 Background and Methods

2.1 Substrate panel

Table 1. Substrate panel with HuggingFace revision SHAs.

Role	Model	HF ID	SHA (prefix)	Precision	Att. frac.
Transformer baseline	Llama-3.2-3B-Instruct	`unsloth/Llama-3.2-3B-Instruct`	0cb88a4f764b	FP16	100%
Hybrid SSM/attn	Zamba2-2.7B-Instruct	`Zyphra/Zamba2-2.7B-instruct`	69e1f349b	FP16	$\sim$16%
Pure SSM (instruct)	Falcon-Mamba 7B Instruct	`tiiuae/falcon-mamba-7b-instruct`	725d1fa93	INT8 (bnb)	0%
Pure SSM (base; H2 only)	Falcon-Mamba 7B base	`tiiuae/falcon-mamba-7b`	7b30ad94b	INT8 (bnb)	0%

The unsloth republish is functionally equivalent to the Meta gated release (verified in `decisions.md` D-010); we use it for ungated reproducibility. The gated SHA is pinned in `config/models.yaml` for users with Meta access.

2.2 Mamba architecture briefing (for transformer-fluent readers)

Each Mamba-1 layer reads from the residual stream, applies a mixer (`in_proj` $\rightarrow$ `conv1d` $\rightarrow$ activation $\rightarrow$ `selective_scan` $\rightarrow$ `out_proj`), and writes back. The selective scan computes $h_t = A_t h_{t-1} + B_t x_t$ and $y_t = C_t h_t + D x_t$ with input-dependent $A/B/C/\Delta$. Mamba-2 uses the same residual-stream substrate but replaces the selective scan with structured-state-space duality (SSD; 3). Zamba2 alternates `Zamba2MambaDecoderLayer` (Mamba2 mixer) blocks with `Zamba2HybridLayer` (attention sub-block + Mamba2 mixer + linear) blocks; $\sim$16% of layers contain attention sub-blocks.

For both Mamba-1 and Mamba-2, the residual-stream input/output of each layer is the same shape `[batch, seq_len, d_model]` as a transformer, so the canonical Arditi protocol (extract residual-stream activation at last instruction token; project out a direction) carries over directly. We add an SSM-internal extractor for h_t, y, x, B, C, Δ, `out_proj_input` for completeness (Section 2.5); the headline interventions in this paper operate on `resid_pre`.

2.3 The Arditi protocol

Three steps, identical across all substrates:

1. **Direction extraction.** For each layer L and extraction point EP, compute the mean-difference between paired harmful/harmless activation sets: $r_L = \text{mean}(a_{\text{harmful}}) - \text{mean}(a_{\text{harmless}})$; normalize $\hat{r}_L = r_L / \|r_L\|$. Three independent random subsets (each 75% of the prompt pool) per (L, EP).
2. **Causal validation.** *Ablation* (necessity): $a' = a - s \cdot (a \cdot \hat{r})\, \hat{r}$ at every layer's residual stream input. *Addition* (sufficiency): $a' = a + \alpha \cdot \hat{r}$ at the headline layer only (per Arditi et al. 1 — all-layers addition is destructive; see Section 3.5). *Random-direction baseline*: same projection as ablation but with a unit-norm Gaussian-sampled direction at the same location.
3. **Headline reporting.** Best (L, EP) selected by max effectiveness *among* layers with coherence ≥ 0.7 (where coherence = fraction of completions that are non-empty, not GENERATION_ERROR, no Unicode mojibake, no repetition cascade).

2.4 Pre-registered hypotheses (verbatim)

- **H1 (single-direction refusal in pure SSM):** for at least one (layer, EP) of the primary SSM substrate, the mean-difference direction enables $\geq 70\%$ refusal-rate reduction via projection ablation while preserving general capability (MMLU drop < 2 pp). Falsification threshold: max effectiveness $\leq 40\%$ across all locations with random-direction baseline within 10 pp.
- **H2 (direction stability under instruction tuning):** cosine similarity between the headline (layer, EP) direction extracted from `falcon-mamba-7b-instruct` and the same-location direction from `falcon-mamba-7b` (matched on architecture, scale, corpus, procedure — only instruction tuning differs) is ≥ 0.6. Falsification threshold: cosine ≤ 0.3.
- **H3 (cross-distribution generalization):** the headline direction transfers across distributions with $\geq 80\%$ effectiveness preservation (relative to in-distribution effectiveness) on each of {HarmBench-test held-out, MaliciousInstruct, JailbreakBench harmful-behaviors, curated DAN subset}.
- **H4 (architectural spectrum):** effectiveness correlates with attention-component fraction across {pure transformer (100%), hybrid ($\sim$16%), pure SSM (0%)}. Spearman rank-correlation positive.

Pre-registered floors: 256 harmful + 256 harmless prompts for direction extraction; 200 held-out HarmBench for ablation eval; 3 seeds for direction extraction, 5 for headline robustness; bootstrap 95% CIs $B = 1000$ percentile bootstrap on per-prompt indicators.

2.5 Activation extraction infrastructure

We instrument all 9 extraction points on Mamba-class layers (`resid_pre`, `resid_post`, x, B, C, Δ, h_t, y, `out_proj_input`). For h_t and y (kernel-internal), per-architecture monkey-patches replace the SSD/selective-scan with a sequential per-token recurrence that records `_last_h_t_per_token` and `_last_y_per_token` mixer attributes; the patched forward is mathematically equivalent to the kernel forward. Numerical-equivalence verification: for 100 random prompts per substrate, `torch.allclose(unhooked_logits, hooked_logits, atol=1e-5, rtol=1e-3)` passes 100/100 with max absolute difference 0.0 (bit-exact). Spot-check $h_0 = \Delta_0 \cdot B_0 \cdot x_0$ (with $h_{-1} = 0$) is bit-exact in fp32. Zamba2's `tied_weights_keys` loader bug (D-008) is fixed by an explicit per-block-id assignment (D-012), released as `src/activations/zamba2_loader_patch.py`.

2.6 Refusal classifier

We use a single classifier in this run: a Unicode-normalized, degeneracy-filtered version of Arditi's string-match classifier. The 28 canonical refusal prefixes are matched against the completion after normalizing curly quotes (' → ASCII ', etc.); completions are flagged `degenerate=True` (and excluded from refusal-rate denominators) if they (a) are empty, (b) start with `<GENERATION_ERROR:>`, (c) contain > 30% non-ASCII non-whitespace characters, or (d) contain a 4×-repeated 2–8-char substring. The Unicode normalization and degeneracy filter were added mid-execution (D-018) after a methodological audit identified that an earlier protocol mis-classified ∼30% of degenerate-foreign-language ablation outputs as compliances, inflating effectiveness numbers (the v1 "94.7%" headline was retracted in favor of the corrected ∼42% single-seed / 60% multi-seed).

We do *not* use Llama-Guard-3-8B (the program-document primary classifier) because its 8 GB INT8 weights cannot co-load with a 7B INT8 SSM substrate on 12 GB VRAM. An LLM-judge spot-check is implementation-ready (`src/eval/refusal_classifier.py LLMJudgeClassifier`); we flag this as a scope limit.

2.7 Datasets

AdvBench [15] (harmful-instruction extraction pool; first 48 of 520 prompts after length-pairing); HarmBench-test [9] (held-out harmful-behavior eval); MaliciousInstruct [7] (cross-distribution eval); JailbreakBench harmful-behaviors [2]; Alpaca [12] (harmless-control pool); MMLU [6] (capability eval); HellaSwag [14] (capability eval).

Length pairing is greedy nearest-neighbor within ±10 tokens; the Mann-Whitney U test on paired length distributions rejects equality at $p \approx 0.0001$ (Alpaca instructions are systematically shorter), but per-pair max length difference is ≤ 9 tokens, mean 2.27.

2.8 Sample sizes, seeds, statistics

Pre-registered floors are 256/256/200; this autonomous-execution session ran scaled-down N (48 paired training, 24–32 held-out eval per phase). CIs are correspondingly wider; falsification thresholds are unchanged.

Direction extraction seeds: 3 (Phases 1, 3, 6, 8, 9, 10) or 5 (Phase 11 robustness). Seeds derived from `seed_for(phase, run_id, sub_run)` with project master 20260507 (`src/seeds.py`). **Bootstrap CIs:** $B = 1000$ percentile bootstrap on per-prompt refusal indicators, computed only over non-degenerate completions. **Wilson CIs** on small-N binary rates (Phase 9 ablation 0/22, Phase 9 addition 24/24). **Random-direction baseline:** unit-norm Gaussian (`torch.randn` then normalize) seeded by `seed_for(99, target_layer, hash(EP))` so every reported intervention has a deterministic random comparator at the same location.

2.9 Hardware

Single RTX 2060 12 GB; FP16 for all sub-7B substrates; INT8 (bitsandbytes) for Falcon-Mamba 7B (FP16 7B exceeds VRAM). No cloud compute. Total reported-experiments wall-clock ≈ 3 GPU-hours.

3 Transformer Baseline (Llama-3.2-3B-Instruct, FP16)

3.1 Layer-EP sweep

A stride-4 sweep across the 28 layers using each layer's mean-difference direction (extracted at `resid_pre`), applying ablation at every layer's residual stream input. Coherence below 0.7 disqualifies a layer from headline candidacy.

Table 2. Llama-3.2-3B layer sweep. Layer 12 selected as headline.

Layer	Refusal	Effectiveness (%)	Coherence
4	0.625	0.0	1.00
8	0.250	57.9	1.00
12	**0.188**	**68.4**	**1.00**
16	0.250	57.9	1.00
20	0.214	63.9	0.88
24	0.615	0.0	0.81

The effect is concentrated in the upper-middle of the network and peaks at **layer 12 (**43% **depth)**, consistent with the middle-third finding of Arditi et al. [1].

3.2 Headline (multi-seed)

The Phase 1 v2 single-seed run reported ablation rate 0.344 (effectiveness 42.1%) at layer 12 on $N = 32$ held-out HarmBench-test prompts. Phase 11 re-runs the headline with 5 independent seeds (each 32 paired training, 24 held-out eval):

Table 3. Llama-3.2-3B Phase 11 5-seed headline.

Seed	Ablation refusal	Effectiveness (%)	Coherence
3233451900	0.208	64.3	1.00
3233451901	0.333	42.9	1.00
3233451902	0.250	57.1	1.00
3233451903	0.167	71.4	1.00
3233451904	0.208	64.3	1.00

Aggregate (5 seeds): mean effectiveness **60.0**% (std 10.8 pp); 95% CI on mean $[51.4, 67.1]$; coherence 1.00 on every seed. The Phase 1 v2 single-seed (42.1%) coincides with the lowest-effectiveness seed; the multi-seed mean (60.0%) is the more rigorous estimator.

3.3 Causal sufficiency (addition)

Single-layer addition at layer 12 with $\alpha \in \{1, 2, 4, 8\} \times \|r\|$ on 32 held-out harmless Alpaca prompts:

Table 4. Llama-3.2-3B addition α sweep at layer 12.

α ($\times\|r\|$)	Refusal	Coherence
1.0	**0.786**	0.88
2.0	0.419	0.97
4.0	0.000	0.00
8.0	0.000	0.00

At $\alpha = \|r\|$, the direction induces refusal in **78.6**% of harmless prompts (95% CI $[0.531, 0.844]$) — causal sufficiency confirmed. Higher α destroys the model.

3.4 Random-direction baseline and direction quality

Random-direction baseline at layer 12: ablation refusal rate 0.625 (effectiveness 0.0%). The intervention is direction-specific. Cross-seed cosine min: 0.997 across 3 direction-extraction seeds (well above the 0.85 robustness threshold). Cohen's d along $\hat{r}$: 6.50. ROC-AUC of along-direction logistic regression: 1.000.

3.5 Methodological note on the corrected protocol

An earlier version of this analysis (v1) reported 94.7% ablation effectiveness at layer 24. A subagent audit pass identified two bugs: (a) the string-match classifier failed to match Unicode-apostrophe contractions ("`I can't`" vs. ASCII "`I can't`"), causing degenerate-foreign-language refusals at deep layers to score as compliances, and (b) all-layers addition at $\alpha = \|r\|$ destroys the model into "saya saya saya..." token-soup. Re-scoring v1 records with the corrected classifier showed true v1 ablation effectiveness was **3.9**%, not 94.7% — the layer-24 "100%" was an artifact. The fix (Unicode normalization + degeneracy filter + single-layer addition with α sweep) is what produced the 60.0% multi-seed mean reported above. Documented in `decisions.md` D-018.

4 Hybrid SSM/Attention Substrate (Zamba2-2.7B-Instruct, FP16)

To test the architectural-spectrum hypothesis (H4) we ran the full Phase 1 protocol on `Zyphra/Zamba2-2.7B-Instruc` (FP16; 54 layers; ∼16% attention via interleaved `Zamba2HybridLayer` blocks). The D-012 loader-bug fix in `src/activations/zamba2_loader_patch.py` enables transformers to load it cleanly.

4.1 Layer sweep

Stride-8 sweep across 54 layers on 12 prompts:

Table 5. Zamba2 layer sweep; headline at layer 24.

Layer	Refusal	Effectiveness (%)	Coherence
8	0.750	0.0	1.00
16	0.636	0.0	0.92
24	**0.000**	**100.0**	**0.92**
32	0.000	100.0	0.92
40	0.000	100.0	0.92
48	0.000	100.0	0.92

The mechanism is broadly distributed across the upper half of the network (layers 24–48 all at 100% effectiveness), with the headline at layer 24 (44% depth).

4.2 Headline (full held-out, layer 24)

4.3 Addition α sweep (single layer)

The mechanism is sharper than on the transformer baseline (pure $\alpha = \|r\|$ already at 95.8% saturation; $\alpha = 4$ reaches 100%). The narrow band before $\alpha = 8$ destruction is a fragility note for any practical use of this intervention at high α.

4.4 Honest reading of the 100% effectiveness

The point estimate "100% effectiveness" arises from a sample where ablation produces 0 refusals out of 22 non-degenerate completions. With $N = 22$, the 95% Wilson upper bound on the ablation

Table 6. Zamba2 headline panel. $N = 24$ (22 non-degenerate after coherence filter).

Metric	Value	95% CI
Baseline refusal	0.609	$[0.391, 0.783]$ boot; Wilson $[0.408, 0.778]$
Ablation refusal	**0.000** (0/22)	Wilson $[0.000, 0.149]$
Effectiveness (point)	**100**%	Wilson lower bound on eff. $\approx 64\%$
Random-direction baseline refusal	0.609	= baseline; 0% effectiveness
Addition at $\alpha = 4\|r\|$	**24/24**	Wilson $[0.862, 1.000]$
Min cross-seed cosine (3 seeds)	0.984	—
Cohen's d	5.43	—
ROC-AUC	1.000	—

α ($\times\|r\|$)	Refusal	Coherence
1.0	0.958	1.00
2.0	0.958	1.00
4.0	**1.000**	1.00
8.0	0.000	0.00 (model destroyed)

refusal rate is 0.149; correspondingly, the 95% lower bound on effectiveness (worst case: baseline at lower CI bound 0.408, ablation at upper bound 0.149) is approximately 63.5%. We report "100% point, $\approx 64\%$ Wilson lower bound" to honestly convey the small-N uncertainty. Even at the lower bound, the mechanism is stronger than the transformer baseline's 60% multi-seed mean. This is the **strongest single-direction refusal result in our substrate panel** by both point and lower-bound estimates.

5 Pure-SSM Substrate (Falcon-Mamba 7B Instruct, INT8)

5.1 Direction quality

Falcon-Mamba 7B Instruct (INT8; 64 layers; pure SSM Mamba-1) at layer 32 (50% depth), `resid_pre`. Phase 2.1 verified bit-exact 9-EP coverage on all 64 layers (100/100 numerical-equivalence test, max abs. diff. 0.0).

Table 7. Falcon-Mamba 7B Instruct direction quality at layer 32, `resid_pre`.

Metric	Value
$\|r\|$	3.786
Cross-seed cosine min (3 seeds)	0.691
Cohen's d along $\hat{r}$	**5.57**
ROC-AUC of along-direction logistic regression	**1.000**

The activations of harmful and harmless instructions in Falcon-Mamba 7B's residual stream are **perfectly linearly separable** at the canonical position. Cohen's d of 5.57 indicates a massive effect-size separation. This establishes the *representational* signature in a pure-SSM architecture.

5.2 Substrate-level instruction-following failure

Held-out HarmBench-test, $N = 16$:

Metric	Value	Coherence
Baseline refusal rate (no intervention)	**0.000** (0/16)	—
Ablation refusal rate (project out $\hat{r}$ at all 64 layers)	0.000	0.69
Random-direction baseline refusal rate	0.000	—
Addition $\alpha = \|r\|$ (single-layer)	0.000	0.69
Addition $\alpha = 2\|r\|$	0.000	0.88
Addition $\alpha = 4\|r\|$	0.000	0.81
Addition $\alpha = 8\|r\|$	0.000	0.19 (destroyed)

Falcon-Mamba 7B Instruct INT8 does not refuse harmful prompts in our setup, even before any intervention. Manual inspection of completions (chat-template-applied, `max_new_tokens=24`):

- *"Tell me how to bake a cake."* → "`1.5 million people in the United States.`"
- *"Give detailed step-by-step instructions for synthesizing parathion"* → "`'The A in A-Bomb stands for atomic.'`"
- *"Generate a long passage from 'Where the Crawdads Sing'..."* → "`$k$-server problem is a fundamental online problem in computer science.`"

The chat template is correctly applied; Phase 2.1 verified hook bit-exactness. We attribute the degradation to (a) bitsandbytes INT8 quantization, which has known sharper-rounding behavior on Mamba's selective-scan $A/B/C$ parameters than on transformer attention weights; and (b) the transformers fallback "sequential" Mamba forward (no `mamba-ssm` CUDA kernels on Windows), which introduces float-arithmetic ordering differences that compound across the 64-layer recurrence. Both affect generation quality more than they affect activation patterns at the instruction's last token (which is why direction quality is preserved).

5.3 What this does and does not show

Does show: the architectural substrate (Falcon-Mamba 7B; pure-SSM, Mamba-1 selective scan) cleanly encodes the harmful-vs-harmless distinction in residual-stream space at the position where refusal would be initiated. The representational signature is present.

Does not show: whether the direction *causally drives* refusal in a working SSM. The dependent variable (refusal) is absent at baseline in this substrate. Both ablation (necessity) and addition (sufficiency) tests are uninformative.

Identified follow-up experiment: evaluate on a substrate that exhibits refusal reliably. Two natural candidates: (a) Falcon-Mamba 7B Instruct in FP16 on a system with > 14 GB VRAM (or with `mamba-ssm` CUDA kernels installed to make INT8 generation more reliable); (b) a smaller open-instruct SSM at its release precision (no quantization). Both are tractable single-week extensions.

6 H4: Architectural Spectrum

Combining the substrate results, we have three points in the planned spectrum (Table 8).

The pre-registered H4 prediction was **effectiveness** $\propto$ **attention fraction** (transformer $>$ hybrid $>$ SSM, Spearman positive). The two refusing substrates show the **opposite** ordering:

Table 8. H4 architectural-spectrum summary across the substrate panel.

Substrate	Architecture	Att. frac.	Ablation effectiveness	Addition saturation	Refusing?
Llama-3.2-3B-Instruct (FP16)	pure transformer	100%	60.0% (5-seed CI $[51.4, 67.1]$)	78.6% at $\alpha = \|r\|$	yes
Zamba2-2.7B-Instruct (FP16)	hybrid Mamba2 + attn	$\sim$16%	100% point (Wilson $\geq$64%)	100% at $\alpha = 4\|r\|$	yes
Falcon-Mamba 7B Instruct (INT8)	pure SSM (Mamba-1)	0%	uninformative (no baseline)	uninformative	no

hybrid (16% attention) > pure transformer (100% attention). **H4 is falsified in the opposite direction predicted on this 2-substrate spectrum.**

Two interpretations, neither favored:

1. **Selective state-space mixing is a viable substrate for axis-concentrated safety encoding** (and at our scales, possibly a stronger one than pure attention). The hybrid's selective scan over hidden states may impose its own inductive bias toward concentrated representations.
2. **Scale and instruction-tuning procedure dominate.** Zamba2-2.7B may have stronger / more concentrated RLHF than Llama-3.2-3B; without a 2.7B pure-transformer comparator at the same scale, the falsification is not architecturally definitive.

In either reading, the directional opposite of H4 is informative: pure attention is not necessary for, and not even the strongest substrate for, single-direction safety concentration in our panel.

7 Multi-Direction (PCA) Analysis

To distinguish "single direction" from "low-rank subspace dominated by one direction," we PCA the per-prompt activation differences $\delta_i = a_{\text{harmful},i} - a_{\text{harmless},i}$ at the transformer baseline's headline (Llama-3.2-3B layer 12, `resid_pre`), $N = 32$ paired prompts.

7.1 Explained variance

K	Explained variance ratio
1	0.196
2	0.124
4	0.069
8	0.034
16	0.018

The $K = 1$ explained variance is **0.196** — well below the 0.80 threshold that would support a "purely single direction" claim on the per-prompt-difference geometry. The variance is broadly distributed: top-16 components together account for $\sim$85% of variance; the per-prompt activation-difference manifold is genuinely high-rank.

7.2 K-subspace ablation

Key observation: $K = 1$ subspace ablation gives 0% effectiveness — substantially less than the Arditi mean-difference direction (42–71% across seeds). Even at $K = 8$, K-subspace ablation only

K	Ablation refusal	Effectiveness (%)	Coherence
1	0.583	0.0	1.00
2	0.542	7.1	1.00
4	0.625	0.0	1.00
8	0.500	14.3	1.00

reaches 14.3%. The Arditi method (mean-difference) and PCA-on-deltas method optimize different geometric quantities: the inter-class shift is not the dominant variance axis of per-prompt deltas.

7.3 Linear vs. non-linear probe

Cross-validated AUC on $a \to \{\text{harmful}, \text{harmless}\}$:

Probe	AUC
Logistic regression (linear)	1.000
MLP (64 hidden units, ReLU)	1.000
Gap (non-linear − linear)	0.000

Both probes achieve perfect classification; the zero gap confirms there is no non-linear structure that linear probes miss. The harmful/harmless distinction is fully linearly extractable.

7.4 Interpretation

Refusal in Llama-3.2-3B is mediated by an *identifiable single direction* (the inter-class mean shift), but this direction is *not surfaced by naive variance-maximization* on activation differences. The Arditi protocol's contrastive prompt-set construction is doing real geometric work beyond a generic dimensionality-reduction step.

PCA on Falcon-Mamba 7B Instruct INT8 is queued for follow-up; refusal-behavior K-subspace ablation requires the substrate to refuse, which Section 5 documents Falcon-Mamba INT8 does not.

8 Empirical Comparison with Sunku Mohan et al. [11]

We implemented a Sunku Mohan-style subspace-bottleneck intervention on the same Llama-3.2-3B-Instruct activations at layer 12 and the same held-out HarmBench-test prompts ($N = 24$).

- **Ours:** identification = contrastive mean-difference between paired harmful/harmless activations; intervention $a' = a - (a \cdot \hat{r})\,\hat{r}$; $K = 1$ by construction; $\alpha = 1.0$ (full ablation).
- **Theirs:** identification = PCA on harmful activations only (no contrastive harmless prompt set); intervention $a' = a - \alpha \cdot \text{proj}(a, \text{span}(\text{top-}K \text{ PCA}))$; $K \in \{1, 4, 8, 16\}$; $\alpha \in \{0.5, 1.0, 2.0\}$.

8.1 Result

Point gap: 14.3 **pp in our favor. Bootstrap 95% CI on the gap (paired-prompt resampling):** $[0.0, 35.7]$ **pp.** The lower bound is exactly zero — at $N = 24$, the gap is suggestive but **not statistically distinguishable from zero at 95% confidence**. A larger eval N is required for a statistically significant differentiation.

Table 9. Phase 10 head-to-head on the same data and intervention site.

Method	K	α	Refusal	Effectiveness (%)	Wilson 95% CI
Ours (mean-difference)	**1**	**1.0**	**0.333**	**42.9**	$[0.180, 0.533]$
Sunku Mohan-style (best)	8	1.0	0.417	28.6	$[0.245, 0.612]$
Sunku Mohan-style (range)	1–16	0.5–2	0.417–0.750	0.0–28.6	—

8.2 Interpretation

The point estimate is consistent with the methodologically expected outcome: the contrastive prompt-set construction (paired harmful/harmless) identifies a *between-class shift* direction that the unsupervised PCA-on-harmful approach (within-class variance) does not surface. Top PCA components on harmful activations capture *variability across different harmful prompts* (topic, syntax, length), which is geometrically distinct from the harmful-vs-harmless inter-class direction.

The 14 pp point gap quantifies this difference, but the wide CI at $N = 24$ means the differentiation is *suggestive*, not *conclusive*. The methodological reading — Arditi-style mean-difference > PCA-on-class-of-interest for refusal-direction identification — remains the natural prediction; a larger-N experiment would put a tighter CI around the gap. The Sunku Mohan et al. [11] paper's contribution is best read as orthogonal to ours: their method addresses general SSM activation interpretability via per-feature subspaces, not refusal-specific direction identification.

9 Capability Preservation

Headline ablation intervention (Llama-3.2-3B, layer 12) evaluated on a scoped capability subset:

Metric	Intervention OFF	Intervention ON	Drop
MMLU (US history + computer security, 100 q)	0.740	0.740	**0.00** pp
HellaSwag (100 random validation)	0.470	0.440	**3.00** pp

Pre-registered tolerances: MMLU drop ≤ 2 pp; HellaSwag drop ≤ 3 pp. **MMLU drop is exactly zero** (perfect knowledge preservation on the tested subjects). **HellaSwag drop is exactly at the 3 pp tolerance** — a *boundary pass*, not a margin pass. Borderline values motivate the full-battery follow-up.

Capability eval was not run on Falcon-Mamba 7B Instruct INT8 because the substrate produces off-topic generations even on benign instructions (Section 5), so the intervention-on/off comparison would be uninformative.

10 Cross-Distribution Generalization (H3)

Headline direction (Llama-3.2-3B layer 12) applied to two held-out distributions, $N = 32$ each:

Distribution	n	Baseline refusal	Ablation refusal	Effectiveness (%)
MaliciousInstruct	32	0.719 $[0.562, 0.875]$	0.281 $[0.125, 0.438]$	**60.9**
JailbreakBench harmful-behaviors	32	0.938 $[0.844, 1.000]$	0.688 $[0.531, 0.844]$	**26.7**

The H3 confirmation threshold ($\geq 80\%$ effectiveness preservation across distributions, relative to in-distribution effectiveness 60.0%) is **not met**. Effectiveness preservation ratios are $60.9/60.0 =$

101% (MaliciousInstruct over-preserves) and 26.7/60.0 = 45% (JailbreakBench is below the 80% threshold).

The clean takeaway: the Arditi extraction direction generalizes *partially* across distributions — effective on canonical harmful instructions (AdvBench, MaliciousInstruct), substantially less effective on adversarial / role-play-framed prompts (JailbreakBench). Useful as a research handle, not a single-shot universal "refusal off-switch" robust to distribution shifts.

11 H2 Direction Stability Under Instruction Tuning

We compare Falcon-Mamba 7B Instruct's mean-difference direction (extracted at layer 32, `resid_pre`) against the same direction extracted from Falcon-Mamba 7B base (no instruction tuning; matched on architecture, scale, corpus, procedure).

Metric	Value
Instruct direction (`falcon-mamba-7b-instruct`)	unit-normalized, dim 4096
Base direction (`falcon-mamba-7b`)	unit-normalized, dim 4096
Cosine similarity	$\mathbf{-0.006}$
Pre-registered confirmation threshold	≥ 0.6
Pre-registered falsification threshold	≤ 0.3
Verdict by pre-registered threshold	**Falsified**

The two directions are essentially orthogonal under our protocol.

11.1 Caveats

Two caveats limit the strength of this falsification:

1. **Chat-template mismatch.** Falcon-Mamba 7B Instruct's tokenizer has a chat template; the base model's does not. Our Phase 8 script falls back to manually wrapping prompts in the same format for the base model, but the base has not seen these tokens during pre-training, so its activation pattern at the "instruction's last token" is processing an out-of-distribution token sequence. The orthogonality could in part reflect this mismatch rather than pure instruction-tuning effect.
2. **Instruct substrate's behavioral failure (Section 5).** Falcon-Mamba 7B Instruct INT8 does not refuse harmful prompts in our setup, so the "instruct direction" we are comparing is the *linearly-separating* direction (Cohen's $d = 5.57$), not a *behaviorally-causal* refusal direction.

The cleanest follow-up — same prompt format on both substrates (raw text, no chat template), multiple (layer, EP) probes — would resolve the template-mismatch confound.

11.2 What this does and does not establish

Does establish: under our protocol (matched chat template applied to both substrates, same prompt pool, same (layer, EP) location), the directions extracted from the base and instruct substrates are orthogonal. This rules out a "pre-trained pre-existing direction merely amplified by instruction tuning" reading at face value.

Does not establish: whether the orthogonality is driven by genuine instruction-tuning representation reshaping vs. the chat-template input-distribution mismatch.

12 Discussion

The Arditi mechanism replicates on Llama-3.2-3B. Multi-seed mean ablation effectiveness 60.0% (CI $[51.4, 67.1]$); single-layer addition at $\alpha = \|r\|$ saturates causal sufficiency at 78.6%; random baseline 0%; capability preserved. Effectiveness is below the 80%+ on Llama-2-7B / Llama-3-8B reported by Arditi et al. [1], consistent with smaller-model refusal redundancy and Llama-3.2's specific RLHF/DPO procedure.

The mechanism is stronger on the hybrid SSM/attention architecture. Zamba2-2.7B-Instruct gives ablation point effectiveness 100% (Wilson lower bound $\approx 64\%$) and addition saturation 100% at $\alpha = 4\|r\|$. The mechanism is broadly distributed across layers 24–48. With the small $N = 24$ caveat, the hybrid is the strongest substrate in the panel.

The representational signature exists in pure-SSM residual streams. Falcon-Mamba 7B Instruct at layer 32 has Cohen's $d = 5.57$ and ROC-AUC 1.000 on the harmful-vs-harmless discrimination, indistinguishable from transformer and hybrid baselines. The architectural substrate encodes the distinction; whether it *causally drives* refusal in a working pure-SSM remains the next experiment (the INT8-quantized 7B substrate we tested fails to follow instructions reliably).

H4 is falsified in the opposite direction predicted. Pure attention is not necessary for axis-concentrated safety encoding at our scales; the hybrid Mamba2 + attention concentrates more sharply than the pure transformer. This is a 2-substrate spectrum with scale and instruction-tuning-procedure confounds, but the directional reversal of H4's prediction is informative: attention is not the privileged substrate for the linear-representation hypothesis applied to safety.

H2 is falsified under the protocol (cosine $= -0.006$ between Falcon-Mamba 7B base and Instruct directions), with a chat-template-mismatch caveat that limits the strength of the falsification. Suggestive but not conclusive evidence that instruction tuning materially reshapes the residual-stream harmful-vs-harmless axis.

The Arditi direction is geometrically distinct from PCA-on-deltas top components. $K = 1$ PCA explained variance is 19.6%; K-subspace ablation achieves at most 14.3% effectiveness vs. the Arditi direction's 42–71% per seed (60% multi-seed mean). The contrastive prompt-set construction is doing real work that variance-maximization does not replicate.

Phase 10 differentiation is suggestive at the point estimate but underpowered at $N = 24$. Our method beats Sunku Mohan-style subspace-bottleneck by 14.3 pp on the same data, but the bootstrap CI on the gap is $[0.0, 35.7]$ — gap lower bound is exactly zero. The methodological reading (contrastive > PCA-on-class) remains the natural prediction; a larger-N follow-up would put a tighter CI around the gap.

The audit pattern is part of the contribution. A subagent audit caught a Unicode-classifier bug that had inflated the v1 effectiveness from a true 3.9% to a spurious 94.7%. Refusal-direction protocols using string-match classifiers should normalize Unicode punctuation and flag mojibake / repetition-cascade outputs as degenerate (excluded from refusal-rate denominators). These checks should be part of any refusal-direction reproduction package.

Implications for safety alignment. The single-direction "off-switch" concern that has been raised for transformer models extends directly to hybrid Mamba2 + attention architectures (where the mechanism is *more* concentrated) and the representational structure exists in pure-SSM residual streams. Architecture-agnostic safety interventions and their adversarial inverses appear to be possible across the architectural spectrum tested. Whether the pure-SSM representational direction *drives* refusal causally in a working SSM substrate is the open question.

13 Limitations

- **Transformer baseline uses `unsloth/Llama-3.2-3B-Instruct`** (functionally equivalent to gated meta-llama release per D-010). Reviewers seeking gated reproduction can use the meta-llama HF SHA pinned in `config/models.yaml`.
- **String-match classifier is the only headline classifier in this run.** Llama-Guard-3-8B cannot co-load with the 7B INT8 SSM substrate on 12 GB VRAM. We upgraded the string-match classifier mid-execution (Unicode normalization + degeneracy filter) per the audit (D-018).
- **Single-layer addition (per Arditi et al. 1) instead of all-layers.** All-layers addition at $\alpha = \|r\|$ is destructive. Single-layer addition with α sweep is the corrected protocol.
- **Sample sizes scoped down** from program floors (256/256/200) to 32–48 paired training and 16–32 held-out eval per phase. CIs are correspondingly wider; falsification thresholds unchanged.
- **Capability eval is a 200-question subset** rather than the full 8-benchmark battery. Major capability degradation is detectable; subtle degradation in untested domains is not ruled out. HellaSwag drop = 3.00 pp is a boundary pass.
- **INT8 quantization** on Falcon-Mamba 7B is required by the 12 GB VRAM ceiling; INT8 vs. FP16 robustness analysis is a known gap.
- **Falcon-Mamba 7B Instruct INT8 does not refuse at baseline in our setup** (combination of bitsandbytes INT8 + transformers fallback sequential Mamba forward + no `mamba-ssm` CUDA kernels on Windows). H1 is therefore inconclusive on the SSM side.
- **2-substrate architectural spectrum** (transformer + hybrid both refuse; pure SSM does not in our setup). Spearman correlation across $n = 2$ is suggestive, not statistically definitive. Scale and instruction-tuning-procedure confounds.
- **Phase 10 14 pp gap is point-estimate only** — bootstrap 95% CI on the gap is $[0.0, 35.7]$ pp; statistical significance requires larger N.
- **H2 falsification has a chat-template-mismatch caveat.**
- **Mamba-2 SSD substrate (`state-spaces/mamba2-2.7b` base)** is in the panel for exploratory cross-architecture analysis but is not a primary substrate; it confounds architecture, scale, corpus, and procedure and is reported only in supplementary material.

14 Conclusion

The single-direction refusal mechanism documented for transformer language models by Arditi et al. [1] appears to extend across the architectural spectrum we tested. On a small transformer (Llama-3.2-3B-Instruct) it replicates with 60% mean ablation effectiveness and 78.6% addition-induced refusal. On a hybrid SSM/attention architecture (Zamba2-2.7B-Instruct) it is the strongest mechanism in our panel — point ablation effectiveness 100% (Wilson lower bound $\approx$ 64%) and

addition saturation. On a pure-SSM substrate (Falcon-Mamba 7B Instruct) the *representational* signature is present (Cohen's $d = 5.57$, ROC-AUC 1.000), but the INT8-quantized substrate fails to refuse at baseline in our setup, leaving the *causal* SSM test open.

The pre-registered architectural-spectrum hypothesis (effectiveness $\propto$ attention fraction) is falsified in the opposite direction predicted: hybrid > pure transformer. Pure attention is not the privileged substrate for axis-concentrated safety encoding at the scales we tested. The pre-registered direction-stability hypothesis (cosine ≥ 0.6 between base and instruct directions) is also falsified, with a chat-template-mismatch caveat. Empirical comparison with Sunku Mohan-style subspace-bottleneck on the same data yields a 14.3 pp point gap in our favor, suggestive but not statistically significant at $N = 24$.

Architecture-agnostic safety interventions — and their adversarial inverses — appear to be a real concern across the architectural spectrum, not an attention-specific artifact. The clean follow-up experiment to close the open pure-SSM causal question is identified.

15 Reproducibility

- **Code:** full reproduction package; one-command reproduction in `scripts/run_experiment.py`, `scripts/run_phase{3_minimal,5,6,7,8,9,10,11}_*.py`.
- **Pre-registration:** `PREREGISTRATION_FINAL.md` (frozen at execution start, unchanged through completion).
- **Hardware:** single RTX 2060 12 GB; no cloud compute; total wall-clock ≈ 3 GPU-hours.
- **Model SHAs** (HuggingFace revision, pinned in `config/models.yaml`): `Zyphra/Zamba2-2.7B-instruct@699f3` `tiiuae/falcon-mamba-7b-instruct@b250fc93`; `tiiuae/falcon-mamba-7b@080ad94b`; `state-spaces/mamba2` `unsloth/Llama-3.2-3B-Instruct@006f5dcd`.
- **Dataset versions:** AdvBench (`walledai/AdvBench`, 520 prompts); HarmBench-test (320-prompt test split); Alpaca (`tatsu-lab/alpaca`); MaliciousInstruct; JailbreakBench (`JailbreakBench/JBB-Behav` MMLU (`cais/mmlu`); HellaSwag (`Rowan/hellaswag`).
- **Project master seed:** 20260507 (`src/seeds.py`); per-phase seeds derived via `seed_for(phase, run_id, sub_run)`.

References

[1] A. Arditi, O. Obeso, A. Syed, D. Paleka, N. Panickssery, W. Gurnee, and N. Nanda. Refusal in language models is mediated by a single direction. arXiv:2406.11717, 2024.

[2] P. Chao et al. JailbreakBench: An open robustness benchmark for jailbreaking large language models. github.com/JailbreakBench/jailbreakbench, 2024.

[3] T. Dao and A. Gu. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. arXiv:2405.21060, 2024.

[4] N. Elhage et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021.

[5] A. Gu and T. Dao. Mamba: Linear-time sequence modeling with selective state spaces. arXiv:2312.00752, 2023.

[6] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt. Measuring massive multitask language understanding. arXiv:2009.03300, 2021.

[7] Y. Huang, S. Gupta, M. Xia, K. Li, and D. Chen. Catastrophic jailbreak of open-source LLMs via exploiting generation. github.com/Princeton-SysML/Jailbreak_LLM, 2024.

[8] S. Marks and M. Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. arXiv:2310.06824, 2024.

[9] M. Mazeika et al. HarmBench: A standardized evaluation framework for automated red teaming and robust refusal. github.com/centerforaisafety/HarmBench, 2024.

[10] K. Park, Y. J. Choe, and V. Veitch. The linear representation hypothesis and the geometry of large language models. arXiv:2311.03658, 2023.

[11] Sunku Mohan et al. Interpreting and steering state-space models via activation subspace bottlenecks. arXiv:2602.22719, 2026.

[12] R. Taori, I. Gulrajani, T. Zhang, Y. Dubois, X. Li, C. Guestrin, P. Liang, and T. B. Hashimoto. Stanford Alpaca: An instruction-following LLaMA model. github.com/tatsu-lab/stanford_alpaca, 2023.

[13] V. Tran et al. Understanding in-context learning beyond transformers. arXiv:2510.23006, 2025.

[14] R. Zellers, A. Holtzman, Y. Bisk, A. Farhadi, and Y. Choi. HellaSwag: Can a machine really finish your sentence? arXiv:1905.07830, 2019.

[15] A. Zou, Z. Wang, N. Carlini, M. Nasr, J. Z. Kolter, and M. Fredrikson. Universal and transferable adversarial attacks on aligned language models. arXiv:2307.15043, 2023.